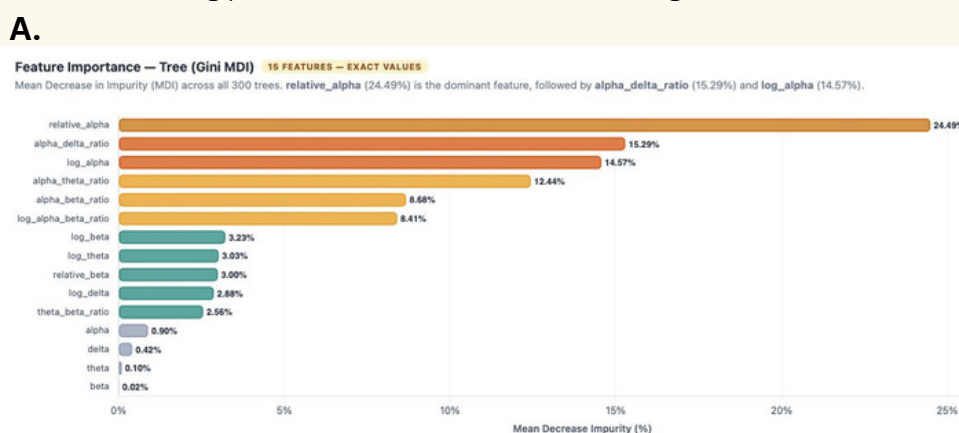


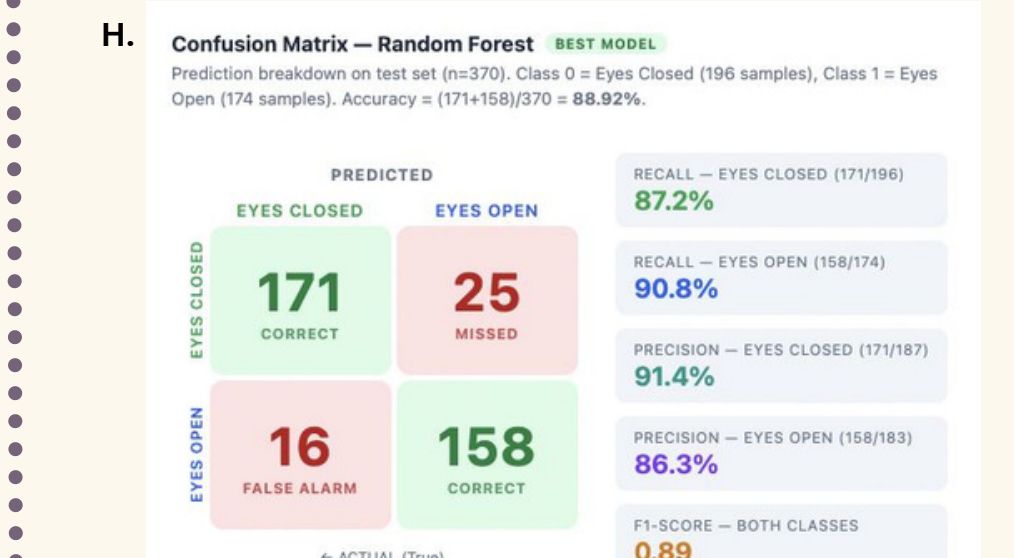
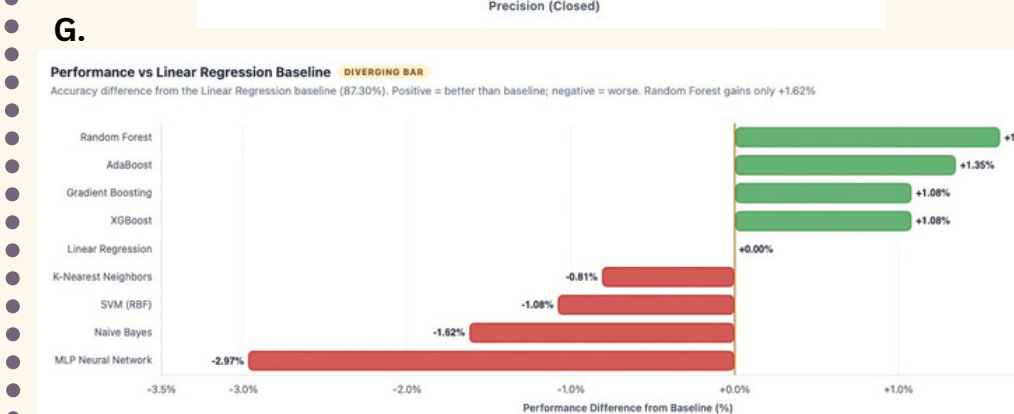
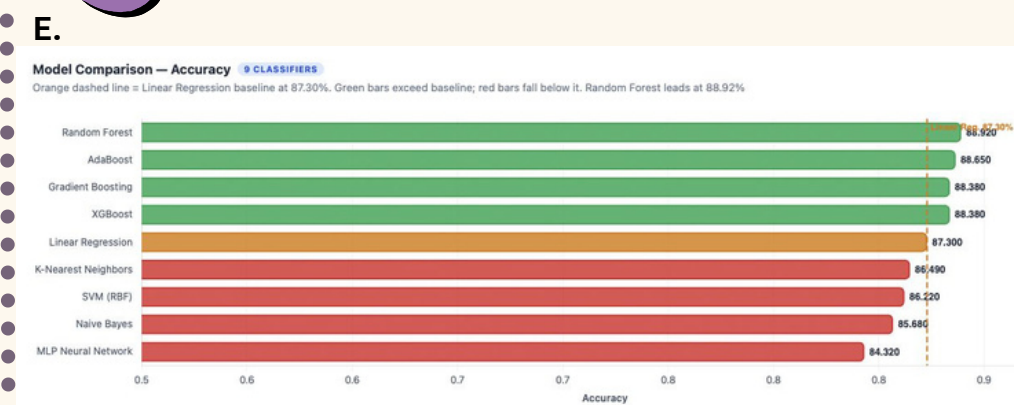


1 Abstract

Machine learning models were trained and evaluated on the extracted features to identify patterns associated with eye-state changes. The results demonstrate that EEG biomarkers can effectively distinguish between open-eye and closed-eye conditions, highlighting the potential of AI-driven approaches for brain signal analysis. Beyond eye-state detection, the developed pipeline provides a foundation for future research into monitoring neurological responses to stimulants and relaxants, where changes in alpha and beta activity serve as measurable biomarkers. This work illustrates how machine learning and EEG analysis can contribute to neuroscience research and emerging applications in drug discovery and neuropharmacology.

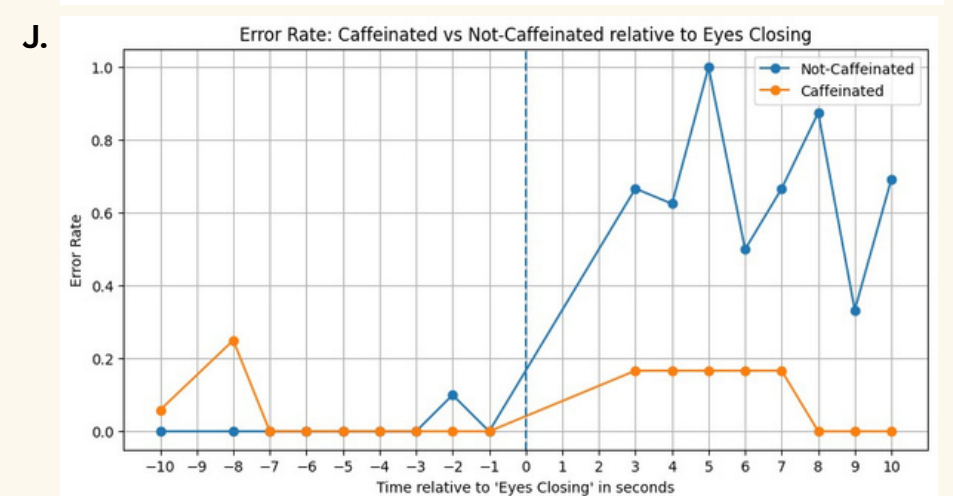
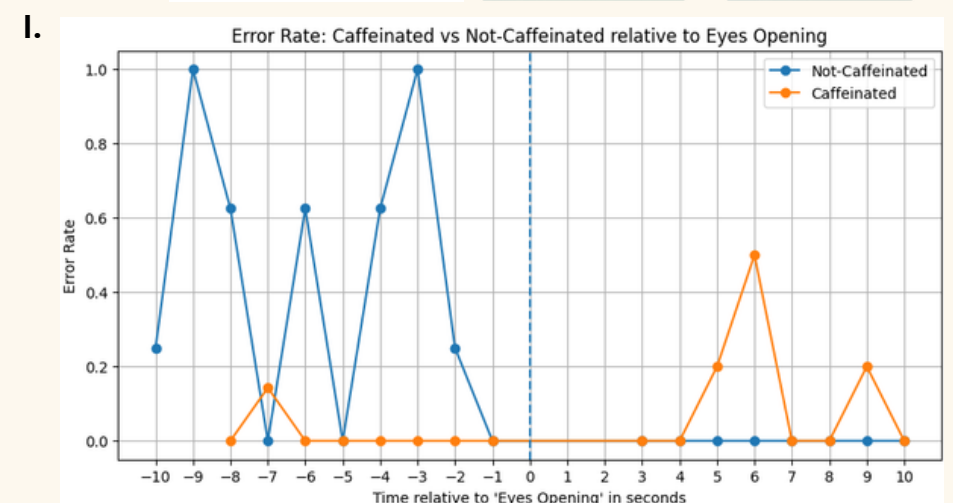


3 Results



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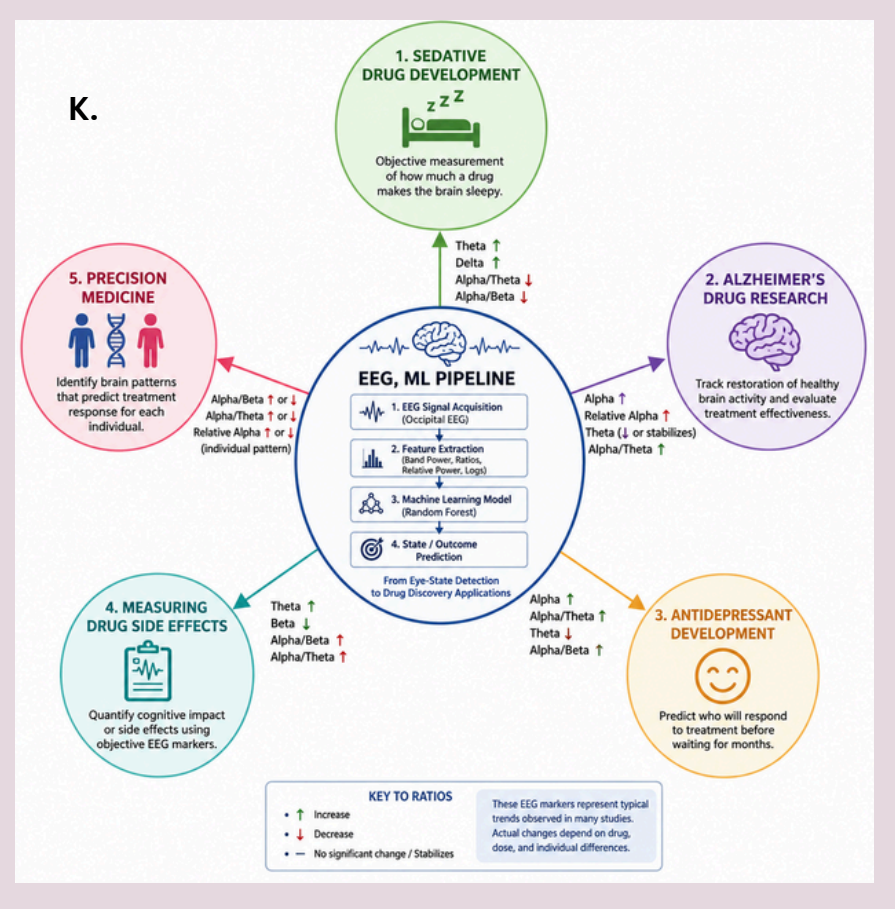
graph TD
    A[Caffeine] --> B[Adenosine Receptor Blockade]
    B --> C[Increased Alertness]
    C --> D[Changes in Alpha/Beta Waves]
    D --> E[Impact on EEG]
    E --> F[Impact on EEG Classification]
  
```



4 Discussion

Among the nine classifiers evaluated, Random Forest achieved the highest accuracy (88.92%), indicating that ensemble learning methods are effective for EEG-based eye-state classification. The model demonstrated balanced performance across both classes, achieving an overall F1-score of 0.89. Feature importance analysis revealed that alpha-related features, particularly relative alpha power and alpha-based ratios, were the strongest predictors of eye state. The model struggles more with eyes-closed EEG data. A possible explanation is that caffeine blocks adenosine receptors (A1/A2A), increasing cortical arousal and altering normal oscillatory patterns. Since eye closure is typically associated with increased alpha activity, caffeine may weaken this relationship, making eyes-closed EEG resemble eyes-open states and reducing classification accuracy. Furthermore, the reliance on alpha activity suggests that similar EEG-based frameworks could be extended to study the effects of stimulants, relaxants, and other neuroactive compounds in future neuropharmacological research, which potentially leads us to drug discovery.

Drug Discovery & Future Applications



References